

Variance Analysis examples.

1. The manager of an offshore wind farm has been asked to produce 150,000 kilowatt hours of electricity in the next year with a budgeted cost of \$1,100 per kilowatt hour. At the request of central management the farm actually produced 160,000 hours. Unit cost was \$1,125 per kilowatt hour. Compute the total variance as well as the price and quantity variances, indicating whether these are favorable or unfavorable. Who should be held accountable for each?
2. For 2023, Mesla, which manufactures and sells electric vehicles, forecasts total revenue of \$7B, based on selling 100,000 vehicles at an average net price of \$70,000 each. At year-end, it reports total revenue of \$6,934,500,000, based on selling 103,500 vehicles. Compute the overall variance as well as the price and quantity variances, indicating whether these are favorable or unfavorable. Who should be held accountable for each?
3. For 2024, Mesla forecasts that it will sell 100,000 units of Model S at an average net price of \$70,000 and 200,000 units of Model 3 at an average net price of \$38,000. At year-end, it reports actual sales of 304,000 vehicles: 74,000 units of Model S and 230,000 units of Model 3, at average net prices of \$72,000 and \$40,000, respectively. Compute the overall variance as well as the price and quantity variances, indicating whether these variances are favorable or unfavorable. Who should be held accountable for each?